

Step1: Problem Definition

Business Setting

Rwanda's textile and apparel sector is expanding gradually, fueled by government measures to promote Made in Rwanda items and minimize reliance on imports. Textile enterprises create fabrics, uniforms, and outfits for both home and international markets. To remain competitive, businesses must watch sales throughout multiple regions, including Kigali, Musanze, Huye, and Rubavu, as well as seasonal demand fluctuations.

Data Challenge:

Sales data is frequently dispersed across multiple countries and months, making it difficult to identify top-selling textile products, assess regional market performance, and estimate customer demand. Without organized analysis, textile companies risk overproducing low-demand goods or missing out on high-demand areas.

The expected outcome is to evaluate sales data from Rwanda's textile sector and provide insights on it's:

- Top selling product per region and per quarter
- Customer purchasing pattern
- Month on month sales growth
- Customer segmentation into purchase power categories
- Forecasting demand with moving averages

This analysis will help Rwanda's textile business improve its production planning, marketing efforts, and resource allocation.

Step 2: Success Criteria

Define exactly 5 measurable goals:

1. Identify the Top 5 textile products per region/quarter → `RANK()`

2. Calculate running monthly sales totals across Rwanda → `SUM()`
`OVER()`
3. Measure month-over-month sales growth → `LAG()` / `LEAD()`
4. Segment Rwandan customers into quartiles based on total spending
→ `NTILE(4)`
5. Compute the 3-month moving average of sales for demand
forecasting → `AVG()` `OVER()`

Screenshots and schemas

data Output Messages Notifications

product_id	name	category	price	stock_quantity
[PK] integer	character varying (100)	character varying (50)	numeric (10,2)	integer

Indexes RLS Policies Rules Triggers employees order_details orders products purchases suppliers

employee_id	name	role	salary
[PK] integer	character varying (100)	character varying (50)	numeric (12,2)
1	Claudine Umutoni	Tailor	150000.00
2	Patrick Nshimiimana	Sales Manager	300000.00
3	Alice Mukandayisenga	Designer	250000.00

Showing rows: 1 to 3 Page 1

order_detail_id	order_id	product_id	quantity	subtotal
[PK] integer	integer	integer	integer	numeric (12,2)
1	1	1	2	16000.00
2	2	2	1	15000.00
3	3	3	3	15000.00
4	4	3	4	9000.00

Showing rows: 1 to 4 Page No: 1 of 1

Total rows: 4 Query complete 00:00:00.139 CRLF Ln 1, Col 1

Data Output Messages Notifications				
Showing rows: 1 to 3 Page No: 1 of 1				
	purchase_id [PK] integer	supplier_id integer	purchase_date date	total_cost numeric (12,2)
1	1	1	2025-09-10	500000.00
2	2	2	2025-09-12	300000.00
3	3	3	2025-09-15	800000.00

✓ Successfully run. Total query runtime: 136 msec. 3 rows affected. ✕

✓ PostgreSQL 13/textile_rwanda - Database connected. ✓

Data Output Messages Notifications					
Showing rows: 1 to 3 Page No: 1 of 1					
	supplier_id [PK] integer	name character varying (100)	contact_person character varying (100)	phone character varying (20)	address text
1	1	Rwanda Cotton Co.	Pauline Mukamana	0788123456	Kigali SEZ, Rwanda
2	2	East Africa Fabrics Ltd.	John Kamali	0788234567	Kampala, Uganda
3	3	Global Textiles Exporters	Li Wei	0788345678	Guangzhou, China

✓ Successfully run. Total query runtime: 123 msec. 3 rows affected. ✕

✓ PostgreSQL 13/textile_rwanda - Database connected. ✕

Total rows: 3 Query complete 00:00:00.123 CRLF Ln 1, Col 1

Data Output

Messages

Notifications

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